Exp no - 13
Date - 12/09/2026
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PACKET SNIFFING USING RAW SOCKETS

AIM:

To capture and display network packets using RAW sockets and identify source and destination MAC addresses, IP addresses, and protocols.

ALGORITHM:

1. Create a RAW socket to capture network packets.
2. Receive packets continuously from the network interface.
3. Extract the Ethernet header information from each packet.
4. Identify the source and destination MAC addresses.
5. Identify the protocol used in the packet.
6. If the packet contains an IPv4 header, extract the source and destination IP addresses.
7. Display the packet details.
8. Continue capturing packets until the user stops the program.
9. Close the socket.

PROGRAM:

```
import socket
import struct

s = socket.socket(
 socket.AF_PACKET,
 socket.SOCK_RAW,
 socket.ntohs(3)
)

print("Packet Sniffer Started...")

try:
 while True:
 packet, _ = s.recvfrom(65535)
```

```
dest, src, proto = struct.unpack(
 "!6s6sH", packet[:14]
 )

mac = lambda x: ":".join(
 "%02x" % b for b in x
 )

print("Source MAC:", mac(src))
 print("Destination MAC:", mac(dest))
 print("Protocol:", hex(proto))

if proto == 0x0806:
 print("Packet Type: ARP")

elif proto == 0x86DD:
 print("Packet Type: IPv6")

elif proto == 0x0800:
 ip = struct.unpack(
 "!BBHHHBBH4s4s",
 packet[14:34]
 )

print("Source IP:",
 socket.inet_ntoa(ip[8]))

print("Destination IP:",
 socket.inet_ntoa(ip[9]))

print()

except KeyboardInterrupt:
 print("Sniffing stopped.")

finally:
 s.close()
```

PROCEDURE:

1. Open a Linux system or virtual machine.
2. Save the program as packet_sniffer.py.

3. Run the program with root privileges using the command: sudo python3 packet_sniffer.py
4. Generate network traffic by browsing websites or using the ping command in another terminal.
5. Observe the captured packet details displayed on the screen.
6. Press Ctrl+C to stop packet sniffing.

OUTPUT:

Packet Sniffer Started...

Source MAC: 00:11:22:33:44:55

Destination MAC: aa:bb:cc:dd:ee:ff

Protocol: 0x0800

Source IP: 192.168.1.10

Destination IP: 192.168.1.1

Protocol: 0x0806

Packet Type: ARP

Sniffing stopped.

Note: The MAC and IP addresses above are illustrative examples. Actual output depends on the network traffic captured.

RESULT:

Thus, packet sniffing was implemented successfully using RAW sockets to capture network packets and display their MAC addresses, IP addresses, and protocols.